

Box Office Intelligence

Data-Driven Recommendations for Launching a New Movie Studio

Team Members

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Executive Summary - Key Takeaways

- ▶ **Budget matters most:** Production budget has the largest effect – +\$3.32 of gross per \$1 spent ($R^2 \sim 0.61$)
- ▶ **Animation is the revenue winner:** Shows a **statistically significant** uplift of ~\$76M after controlling for budget
- ▶ **Ratings vs revenue tradeoff:** Drama/History/War score highest on ratings; Horror scores much lower (ratings harder to predict, $R^2 \approx 0.07$)

These findings should drive our initial portfolio: prioritize high-budget animation with complementary genres for a balanced slate

Business Context & Objectives

Business Context

- ▶ Launching studio with limited budget and zero track record
- ▶ Need genre and budget guidance for early greenlights
- ▶ Portfolio strategy: big bets + prestige arm

Key Questions

- ▶ Which genres generate highest box office revenue?
- ▶ How do budgets interact with genres?
- ▶ Which genres deliver the best revenue–rating balance?

Goal: Identify genres that are both revenue-generating and well-rated to inform greenlighting decisions

Data Approach & Key Model Results

Data & Methods

- ▶ Merged movie dataset: budget, gross, ratings, genres
- ▶ OLS models: Revenue ~ budget + genres; Ratings ~ genres
- ▶ Controlled for budget to isolate genre effects

Revenue Model Results

$R^2 = 0.608$ (explains ~61% of revenue variance)

Budget effect: +\$3.32 gross per \$1 spent

Animation bonus: +\$76M revenue uplift

Ratings Model $R^2 = 0.071$

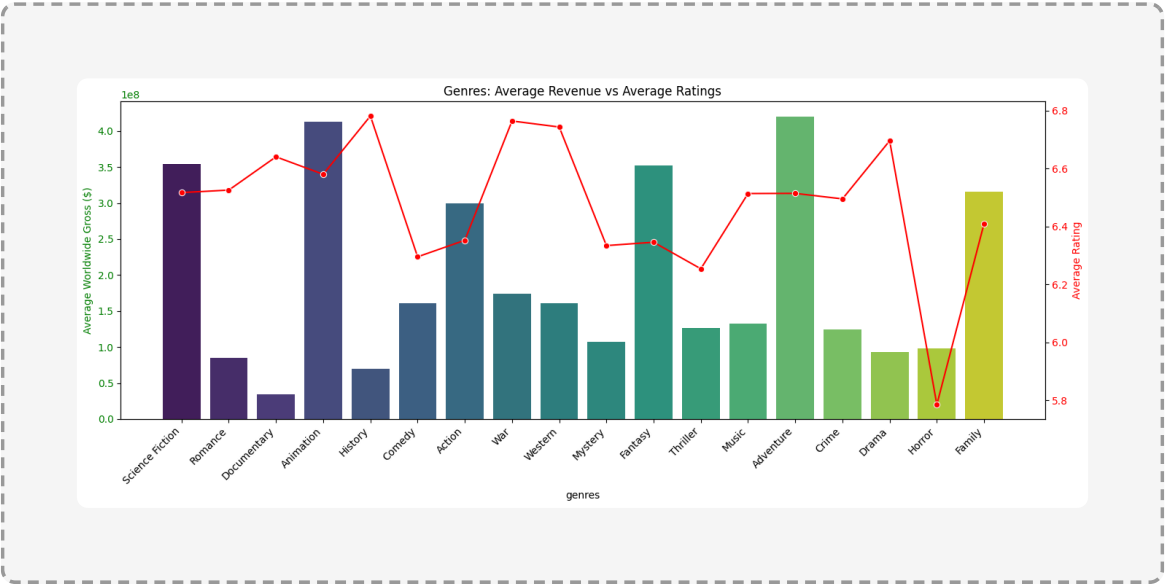
Top: History (+0.43), Drama (+0.34)
Bottom: Horror (-0.57)

Key Insight: Budget drives revenue; genres affect both revenue and reputation differently

Revenue vs Ratings by Genre

Chart: Combined Revenue vs Ratings Analysis

Animation, Adventure, and Science Fiction show strong performance on both revenue and ratings dimensions

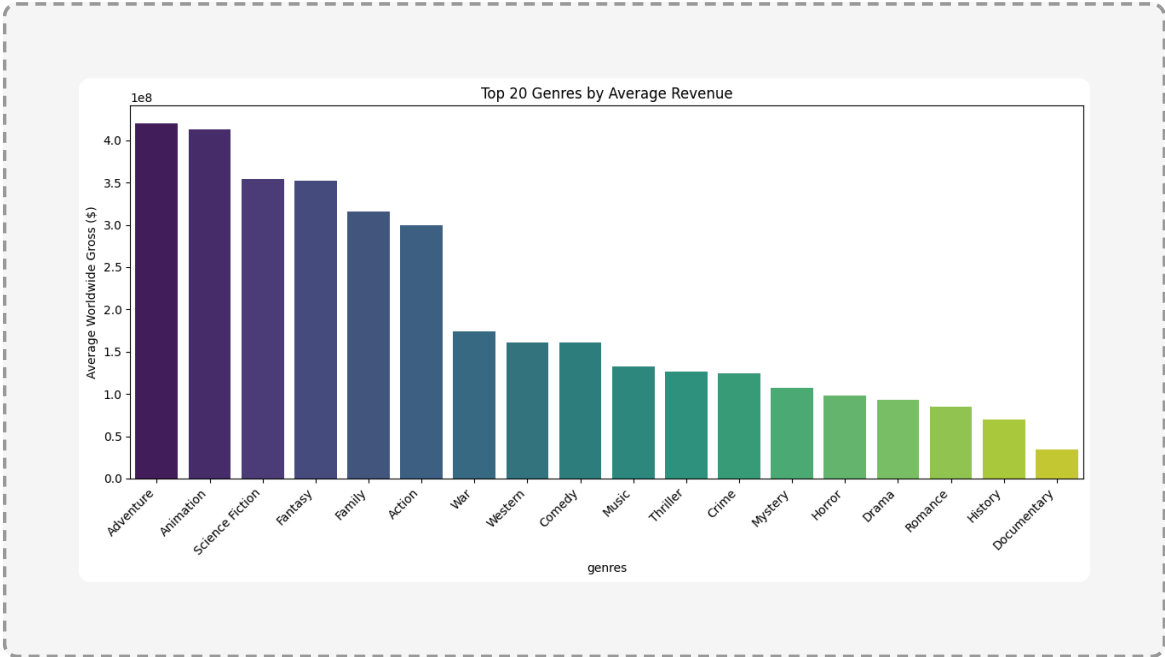


Animation, Adventure, Science Fiction show strong performance on both revenue and ratings

Top Revenue vs Top Rated Genres

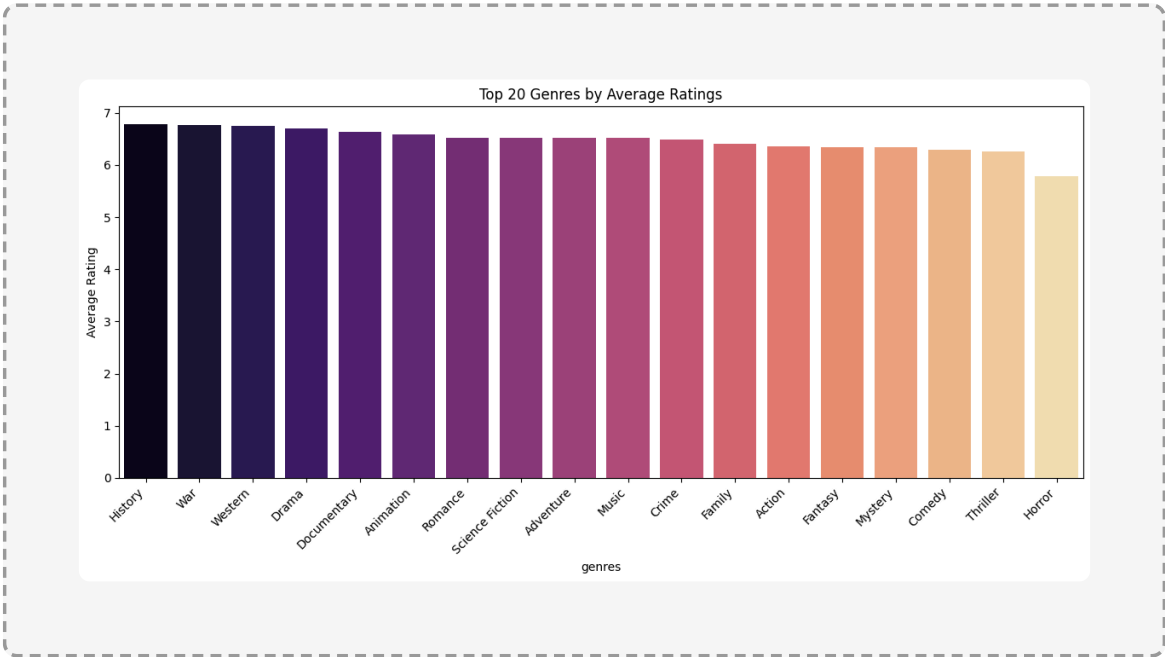
Top Revenue Genres

Adventure, Animation, Sci-Fi



Top Rated Genres

History, War, Drama



Strategic Choice: Revenue sweet spots vs reputation builders – portfolio should balance both

Our Strategic Recommendations

Portfolio Strategy

1. Primary Focus: Animation

Greenlight 1-2 high-budget Animation pilots (high revenue uplift + solid ratings)

2. Secondary Focus: Adventure & Science Fiction

Invest in mid-to-high budgets with franchise potential

3. Reputation Arm: Drama/History/War

Produce 1 prestige film per year for brand credibility

4. Avoid: Horror as Core Pillar

Treat as low-cost experimental vertical only (poor ratings impact)

Implementation Plan & Next Steps

Immediate Actions

Pilot Approval: 1 Animation + 1 Adventure/Sci-Fi + 1 prestige Drama

Budget Envelope: Allocate high budgets for Animation/Adventure pilots

Follow-up Research

- ▶ Run scenario models for pilot scripts at proposed budgets
- ▶ Improve models: log revenue, add controls (stars, marketing, franchise)
- ▶ A/B test Animation concepts with early metrics

Expected ROI: Based on models, high-budget Animation projects show 3.3x budget return + \$76M genre bonus

Limitations & Model Diagnostics

Key Limitations

Causation vs Correlation: Models show association, not causation

Technical Issues: Large condition number suggests multicollinearity

Missing Variables: Star power, marketing spend, franchise status not included

Sample Size: Genre representation varies significantly

Revenue Distribution: Highly skewed by blockbusters

Ratings Predictability: Low R^2 (0.071) indicates many unmeasured factors

Mitigation Strategy: Use models as directional guidance, validate with pilot testing and improved data collection

Ask & Next Steps

Leadership Approval Needed

Pilot Budget Approval: Fund 3-film initial slate (Animation + Adventure/Sci-Fi + Drama)

Research Investment: Approve enhanced data collection and modeling improvements

Timeline: Begin pilot development within 60 days, enhanced models within 90 days

Bottom Line: Prioritize high-budget Animation and Adventure/Science-Fiction for revenue maximization, while maintaining prestige slate for brand reputation. Models show clear \$3.32 return per budget dollar, with Animation providing additional \$76M uplift.